

RRC-Raptor-V4H MSIOF

- User's Manual: Software

V0.0.3

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i. VERSION LIST

Description	Version	Date	Author
RTX_MSIOF	V0.0.1	2024/07/03	Jerry
RTX_MSIOF	V0.0.2	2024/10/29	Jerry
Translate the content into English.	V0.0.3	2024/11/20	Jerry

ii. HASHTAG

#V4H

#MSIOF

#SPI

#BSP

1. Overview

Target : Introduce how to modify the Device Tree and how to use the spidev-test tool to read and write SPI devices.

2. Directory Configuration

The directory configuration is shown below.

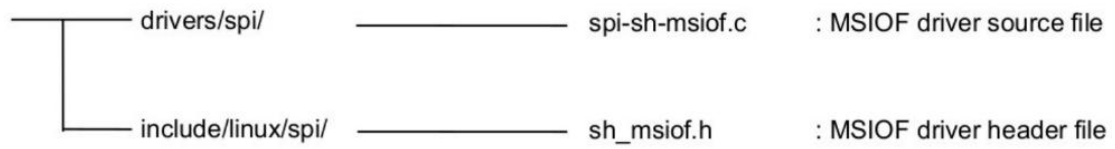


Figure 1. Directory Configuration

3. Pin Function setting

The editing contents are shown in Figure 2 (Example for r8a779g0-raptor.dts)

```
&pfc {  
    msiof1_pins: msiof1 {  
        groups = "msiof1_clk", "msiof1_rxd", "msiof1_txd", "msiof1_sync", "msiof1_ss1", "msiof1_ss2";  
        function = "msiof1";  
    };  
  
    msiof2_pins: msiof2 {  
        groups = "msiof2_clk", "msiof2_rxd", "msiof2_txd", "msiof2_sync", "msiof2_ss1", "msiof2_ss2";  
        function = "msiof2";  
    };  
};
```

Figure 2. Pin Function setting

4. Channel Node setting

Enable MSIOF1 status property for R-Car V4H Raptor.

```
&msiof1 {
    pinctrl-0 = <&msiof1_pins>;
    pinctrl-names = "default";
    num-cs = <3>;

    status = "okay";

    slavedev1@0 {
        compatible = "maker,slavedev";
        reg = <0>;
        spi-max-frequency = <16666666>;
        spi-cpha;
        spi-cpol;
    };

    slavedev1@1 {
        compatible = "maker,slavedev";
        reg = <1>;
        spi-max-frequency = <16666666>;
        spi-cpha;
        spi-cpol;
    };

    slavedev1@2 {
        compatible = "maker,slavedev";
        reg = <2>;
        spi-max-frequency = <16666666>;
        spi-cpha;
        spi-cpol;
    };
};
```

Figure 3 Channel Node setting

Enable MSIOF2 status property for R-Car V4H Raptor.

```
&msiof2 {
    pinctrl-0 = <&msiof2_pins>;
    pinctrl-names = "default";
    num-cs = <3>;

    status = "okay";

    slavedev2@0 {
        compatible = "maker,slavedev";
        reg = <0>;
        spi-max-frequency = <16666666>;
        spi-cpha;
        spi-cpol;
    };

    slavedev2@1 {
        compatible = "maker,slavedev";
        reg = <1>;
        spi-max-frequency = <16666666>;
        spi-cpha;
        spi-cpol;
    };

    slavedev2@2 {
        compatible = "maker,slavedev";
        reg = <2>;
        spi-max-frequency = <16666666>;
        spi-cpha;
        spi-cpol;
    };
};
```

Figure 4 Channel Node setting

5. Kernel configuration

To enable the function of this module, make the following setting with Kernel Configuration. The changes affect after re-building the kernel.

```
Device Drivers --->
  [*] SPI support -->
    --- SPI support
    <*> SuperH MSIOF SPI Controller
    [*] Transfer Synchronization Debug support for MSIOF
    (1) Master of sleep latency (msec time)
        ...
        *** SPI Protocol Masters ***
    <*> User mode SPI device driver support
        ...
    [ ] SPI slave protocol handlers *1
```

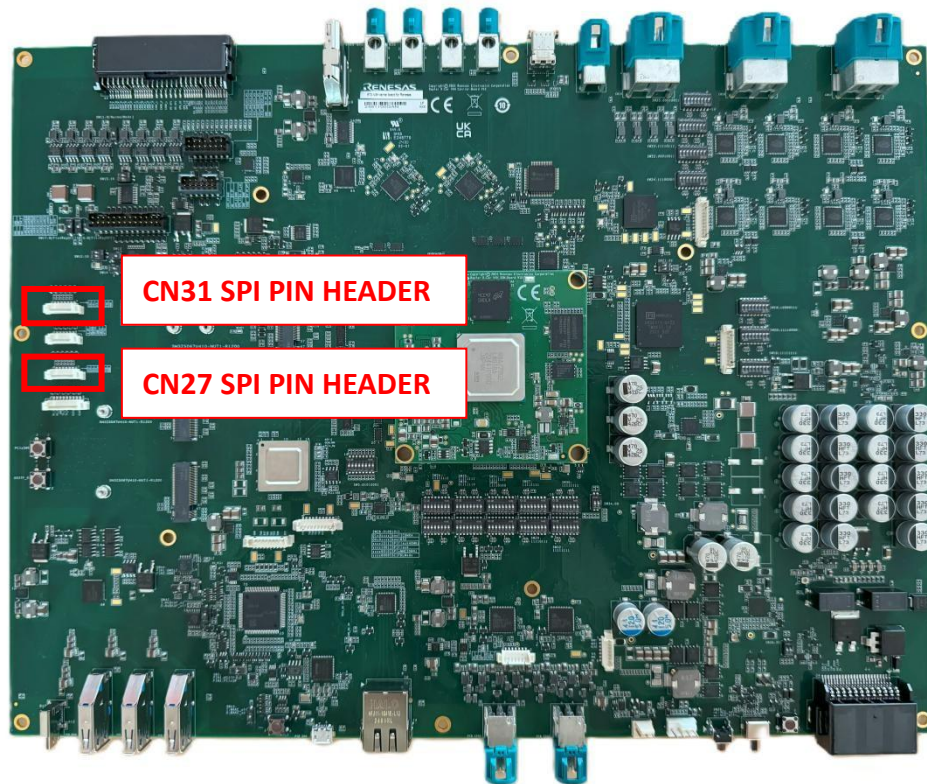
Figure 5 Kernel configuration

Notes:

*1: Please set if you want to enable SPI Slave mode.

6. Connect

Provide an MSIOF interface so that users can add their own SPI devices.



CN31 : SPI PIN HEADER (**spidev0**)

PIN	SYMBOL
1	MSIOF1_SS2/MD18_SW (CS2)
2	MSIOF1_SS1/MD16_SW (CS1)
3	MSIOF1_SCK/MD14_SW (SLK)
4	MSIOF1_SYNC/MD32_SW (CS0)
5	MSIOF1_RXD/MD11_SW (DO)
6	MSIOF1_TXD/MD13_SW (DI)
7	GND

PIN	SYMBOL
1	MSIOF2_SS2/MD18_SW (CS2)
2	MSIOF2_SS1/MD16_SW (CS1)
3	MSIOF2_SCK/MD14_SW (SLK)
4	MSIOF2_SYNC/MD32_SW (CS0)
5	MSIOF2_RXD/MD11_SW (DO)
6	MSIOF2_TXD/MD13_SW (DI)
7	GND

7. SPI Tool

Use spidev-test can read / write spi device.

- I. Here is an example of Raptot connecting to SPI Flash through msiof1 (CN 31).

```
root@v4x:~# spidev_test -HOv -D /dev/spidev0.0 -p '\x90\xff\xff\x00\x00\x00'
spi mode: 0x3
bits per word: 8
max speed: 500000 Hz (500 kHz)
TX | 90 FF FF 00 00 00 |.....|
RX | 00 00 00 00 EF 15 |.....|
```

Figure 6 example of spidev-test method

- II. If using MSIOF2 (CN27), modify the command to:

spidev_test -HOv -D /dev/spidev1.0 -p '\x90\xff\xff\x00\x00\x00'

8. APPENDIX